MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 5727

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, xenon gas
Outline --- 5-2 (EIA)
Cathode --- Coated unipotential
Mounting position --- Any
Weight --- 0.3 ounce (8.5 grams) nominal

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	g1	k	h	h	g2	a	g2

ABSOLUTE RATINGS:

Parameter:	Ef	epp	epy	epx	Conduction Ec2	Conduction Ec1	Nonconduction ec2	Nonconduction ec1	ehk	Rg1	Rg2	ik
Unit:	V	v	v	v	V	V	v	v	v	Meg	Ohms	a
Relay, G controlled rectifier:												
Maximum:	6.9	---	650	1,300	---	---	---	---	25	10	---	0.5
Minimum:	5.7	---	---	---	-10	-10	-100	-100	-100	---	---	---
Pulse modulator:												
Maximum:	6.9	500 (see note 1)	500	100	---	---	---	---	0	0.5	25K	10
Minimum:	6.0	---	---	---	-10	-10	-50	-100	0	---	2K	---
TEST CONDITIONS:	6.3	---	---	---	0	---	---	---	---	---	---	---

ABSOLUTE RATINGS:

Parameter:	Ik	ic2	icl	Ig2	Igl	tp	dik/dt	prr	Du	TA	tk
Unit:	mA (see note 2)	ma	ma	mA (see note 2)	mA (see note 2)	µs	a/µs	---	---	°C	sec
Relay, G controlled rectifier:											
Maximum:	-100	---	---	10	10	---	---	---	---	90	---
Minimum:	---	---	---	---	---	---	---	---	---	-75	20
Pulse modulator:											
Maximum:	10	20	20	---	---	5	100	500	0.001	90	---
Minimum:	---	---	---	---	---	---	---	---	---	-75	20
TEST CONDITIONS:	---	---	---	---	---	---	---	---	---	---	20 (min)

GENERAL:

Qualification - Required
Reliable tube
Holding period (MIL-STD-1311) - t = 24 hours

* U.S. GOVERNMENT PRINTING OFFICE: 1981 — 703-023/1654

FSC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 1							
1201	Short and discontinuity detection		0.4	II	---	---	---	---
3241	Heater current		0.4	II	If	550	650	mA
1336	Heater-cathode leakage	Ehk = +25 Vdc	0.4	II	Ihk	---	15	µAdc
		Ehk = -100 Vdc			Ihk	---	15	µAdc
3201	Critical grid voltage for conduction (1)	Epp = 460 Vac; Rgl = 0.1 Meg; Rp = 3,000 ohms (see notes 3, 5 and 6)	0.4	II	Eccl	-2.9	-4.5	Vdc
3201	Critical grid voltage for conduction (2)	Epp = 460 Vac; Rgl = 10 Meg; Rp = 3,000 ohms (see notes 5 and 6)	0.4	II	Eccl	---	-5.2	Vdc
3201	Critical anode voltage for conduction (1)	Eccl = 0; Rgl = 0.1 Meg; Rp = 1,000 ohms (see note 5)	0.4	II	Ebb	---	38	Vdc
3201	Critical anode voltage for conduction (2)	Ef = 0; Eccl = -100 Vdc; Rgl = 0; Rp = 10,000 ohms (see notes 5 and 8)	0.4	II	Ebb	650	---	Vdc
1231	Pulse emission	e = 180 ±9 v; t = 3 sec (max); Rp = 10 ohms; Ra = 15 ohms; Zm = 7.5 ohms; prr = 100 ±5; calibrating resistor = 5 ohms	0.4	II	etd	---	76	v
---	Operation	Ebb = 500 Vdc (approx); egy = 100 v; epy = 1,000 v; Eccl = -50 Vdc; Ecc2 = 0; prr = 500; Zo = 25 ±5 percent; Rℓ = 20 ±5 percent (see note 9)	0.4	II	ib	16	---	a
	Quality conformance inspection, part 2							
1211	Insulation of electrodes	Ef = 6.3 V; Eg2-p = ±380 Vdc (see note 10)	2.5	S3	R	760	---	Meg
1121	Base strain		---	---	---	---	---	---
2126	Envelope strain	See note 3	2.5	I	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
1031	High-frequency vibration	No voltages applied; fixed amplitude = 0.0625 ±0.0025	6.5	Code E	---	---	---	---
1041	Shock	720 G	6.5	See note 7	---	---	---	---
---	Shock-test end points:							
1336	Heater-cathode leakage	Ehk = +25 Vdc	---	---	Ihk	---	40	µAdc
		Ehk = -100 Vdc			Ihk	---	40	µAdc
3201	Critical anode voltage for conduction (1)		---	---	Ebb	---	50	Vdc
1231	Pulse emission		---	---	etd	---	76	v
3201	Critical grid voltage for conduction (1)		---	---	Eccl	-2.9	-4.5	Vdc
3201	Critical grid voltage for conduction (3)	Ef = 7.0 V; Epp = 460 Vac; Rgl = 10 Meg; Rp = 3,000 ohms (see notes 5, 6, and 11)	---	---	Eccl	---	-6.4	Vdc
3201	Critical anode voltage for conduction (3)	Ef = 5.7 V; Eccl = 0; Rgl = 0.1 Meg; Rp = 1,000 ohms (see notes 5 and 12)	2.5	II	Ebb	---	50	Vdc
1261	Electrode voltage (grid No. 2)	Epp = 150 Vac; Eggl = 16 Vac; Rp = 1,000 ohms; Rgl = 2,500 ohms (see notes 13 and 14)	---	---	Egg2	1.9	3.3	Vac

METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3					
1506	Heater-cycling life	Ef = 7.5 V; Ehk = -100 Vdc; Ec1 = Ec2 = Eb = 0 (see note 15)	---	---	---	---
---	Heater-cycling life-test end point:					
1336	Heater-cathode leakage	Ehk = +25 Vdc	Ihk	---	20	µAdc
		Ehk = -100 Vdc	Ihk	---	20	µAdc
1516	Stability life	Epp = 460 Vac; Rp/Ib = 80 mAdc (min); Rgl = 50,000 ohms; Rp/ib = 500 ma (min); TA = room; tk = 20 +0 -1 (see notes 5 and 16)	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Quality conformance inspection, part 3 -Continued					
---	Stability life-test end points: (2 and 20 hours)					
3201	Critical grid voltage for conduction (1)		Eccl	-2.8	-4.6	Vdc
---	Change in grid voltage (1) of individual tubes		ΔEccl	---	15	%
1501	Intermittent life	Epp = 460 Vac; Rp/Ib = 80 mAdc (min); Rgl = 50,000 ohms; Rp/ib = 500 mA (min); TA = room; tk = 20 +0 -1 Ehk = -100 Vdc; TE = 150°C (min) (see notes 5, 16, and 17) Group E	---	---	---	---
---	Intermittent life-test end points: (500 hours)	See note 19				
---	Inoperatives		---	---	---	---
3241	Heater current		If	550	660	mA
3201	Critical grid voltage for conduction (1)		Eccl	-2.0	-4.8	Vdc
3201	Critical anode voltage for conduction (1)		Ebb	---	50	Vdc
1231	Pulse emission		etd	---	100	v
1336	Heater-cathode leakage	Ehk = +25 Vdc	Ihk	---	20	μAdc
		Ehk = -100 Vdc	Ihk	---	20	μAdc
1211	Insulation of electrodes	g2 = p	R	380	---	Meg
---	Total defectives		---	---	---	---
---	Intermittent life-test end points: (1,000 hours)	See note 19				
---	Inoperatives		---	---	---	---
3241	Heater current		If	550	666	mA
3201	Critical grid voltage for conduction (1)		Eccl	-1.6	-4.8	Vdc
3201	Critical anode voltage for conduction (1)		Ebb	---	120	Vdc
1336	Heater-cathode leakage	Ehk = +25 Vdc	Ihk	---	50	μAdc
		Ehk = -100 Vdc	Ihk	---	50	μAdc
1211	Insulation of electrodes	g2 = p	R	300	---	Meg
---	Total defectives		---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Quality conformance inspection, part 3 -Continued					
---	Life test (continuous)	Group A; epy = 500 v (approx); egy = 100 v (max); Ecc1 = -50 Vdc; Ecc2 = 0; Zo = 12.5 ±5 percent; Rℓ = 7.5 ±5 percent (see note 18)	---	---	---	---
---	Life-test end points: (200 hours) (continuous)					
1231	Pulse emission		etd	---	100	v
---	Operation under continuous life-test conditions	See note 4	ib	16	---	a
1031	Vibration-fatigue	See note 7	---	---	---	---
---	Vibration-fatigue test end points:					
1336	Heater-cathode leakage	Ehk = +25 Vdc	Ihk	---	40	µAdc
		Ehk = -100 Vdc	Ihk	---	40	µAdc
3201	Critical anode voltage for conduction (1)		Ebb	---	50	Vdc
1231	Pulse emission		etd	---	76	v
3201	Critical grid voltage for conduction (1)		Eccl	-2.9	-4.5	Vdc

NOTES:

1. After completion of a pulse, a 20 µs delay is required before a positive voltage of more than 10 v is applied to the tube.
2. Averaged over any interval of 30 seconds maximum.
3. This test shall be performed at the conclusion of the holding period.
4. Thirty seconds shall be the maximum time under test conditions before reading ib. There shall be no evidence of amplitude jitter.
5. Connect pins No. 5 and 7 to pin No. 2.
6. Use miniature steatite socket with grounded shield base. Shield the anode power supply. Use short shielded anode and grid leads. Anode and grid resistors shall be noninductive. Connect the grid resistor directly at socket.
7. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, AQL 6.5, sample size code letter E, shall apply.
8. No voltages shall be applied to the tube for 20 minutes minimum preceding this test.
9. Thirty seconds maximum in the operation test socket is permitted before reading. There shall be no evidence of amplitude jitter. The tube shall be tested in the circuit shown on figure 1. The circuit constants shall be so chosen that: at epy = 1,000 v under resonant charging conditions, dik/dt = 100 a/µs minimum, tp = 2.0 µs ±10 percent, prr = 500. The grid pulse characteristic at egy = 100 v shall be: tp = 2.0 µs maximum, tr = 0.5 µs minimum, driver impedance = 500 ohms minimum.
10. Read electrode insulation between g2 and anode with all other elements floating.
11. Preheat for 15 minutes under the following conditions: Ef = 7.0 V; Epp = 220 Vac; Ecc1 = Ecc2 = 0; Rg1 = 10 Meg; Ib = 100 mAdc. Two seconds shall be the maximum time between preheat and test.
12. Preheat using Ef = 5.7 V.

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NOTES: -Continued

13. Egg1 supply shall be in phase with Epp supply and Egg2 supply 180 degrees out of phase with Epp supply.

14. *Vary Egg2 supply and read Egg2 at which conduction occurs.*

15. The no-load to steady-state full-load regulation of the heater-voltage supply shall not be more than 3.0 percent. This test shall be made on a lot-by-lot basis.

16. Phase of grid voltage adjusted to provide start of conduction at peak applied anode voltage.

17. Envelope temperature (TE) is defined as the highest temperature indicated when using a thermocouple of No. 30 B&S or smaller diameter elements welded to a ring of 0.025-inch diameter phosphor bronze in contact with the envelope. The envelope temperature requirement will be satisfied if a tube, which draws 80 mAdc (±5 percent) anode current and 500 ma (±5 percent) peak anode current under the life-test conditions, is determined to operate at, or above, the minimum specified temperature at any position on the life-test rack.

18. Adjust epy for ib = 20a initially, and maintain this epy value throughout the life test. The tube shall be tested in the circuit shown on figure 1. The circuit constants shall be so chosen that: at epy = 500 v under resonant charging conditions, dik/dt = 100 a/µs minimum, tp = 2.0 µs ±10 percent, prr = 1,000. The grid pulse characteristic at egy = 100 v shall be: tp = 2.0 µs maximum, tr = 0.5 µs minimum, driver impedance = 500 ohms minimum.

19. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples.

20. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 99
DLA - ES

User activities:
Army - AR, ME
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3234)

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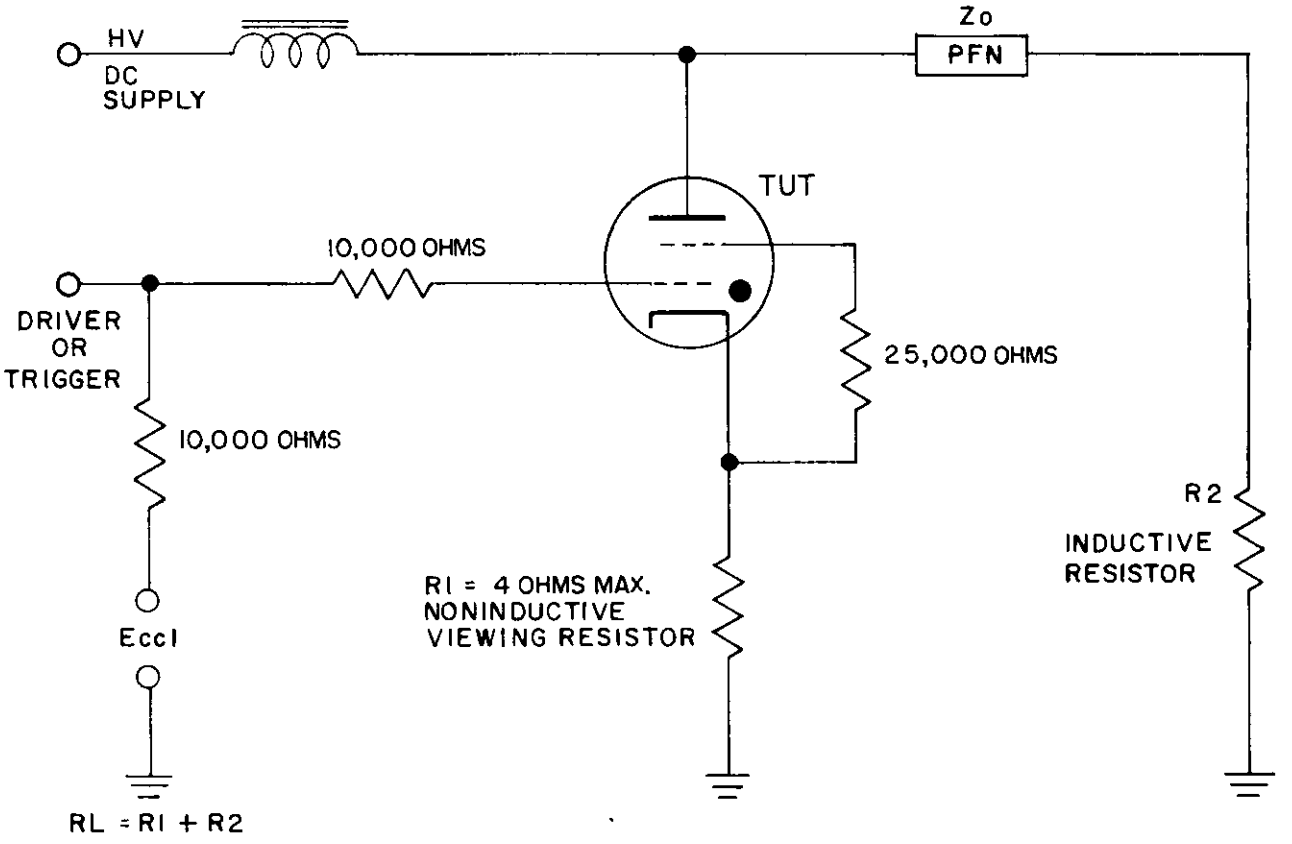


FIGURE 1. Test circuit.

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